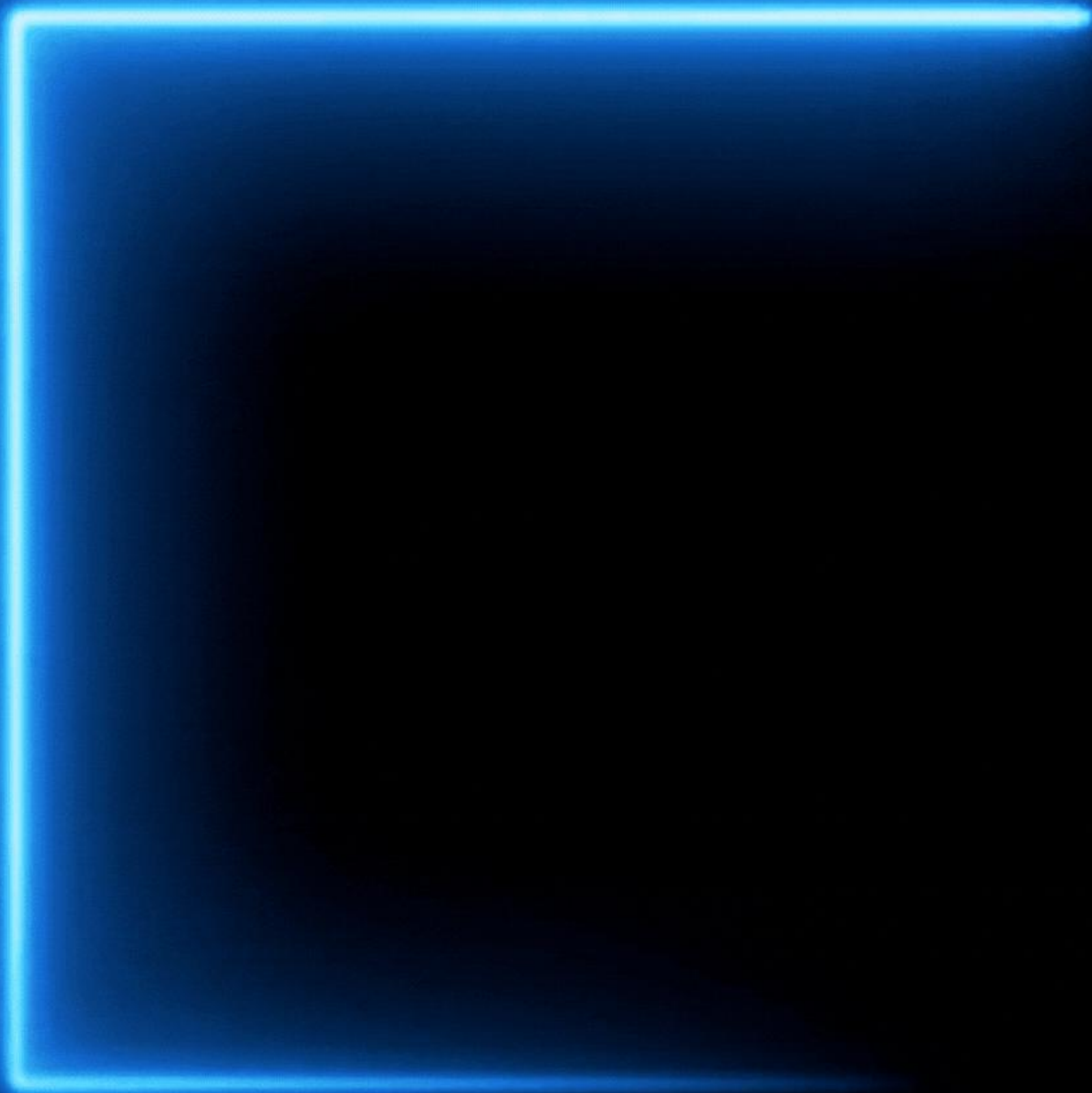




OPERATING SYSTEM

- is the software that manages a computer's hardware and software resources, providing a stable and consistent way for applications to interact with the hardware. It acts as an intermediary between users and the computer hardware, handling tasks such as memory management, process scheduling, and input/output operations.



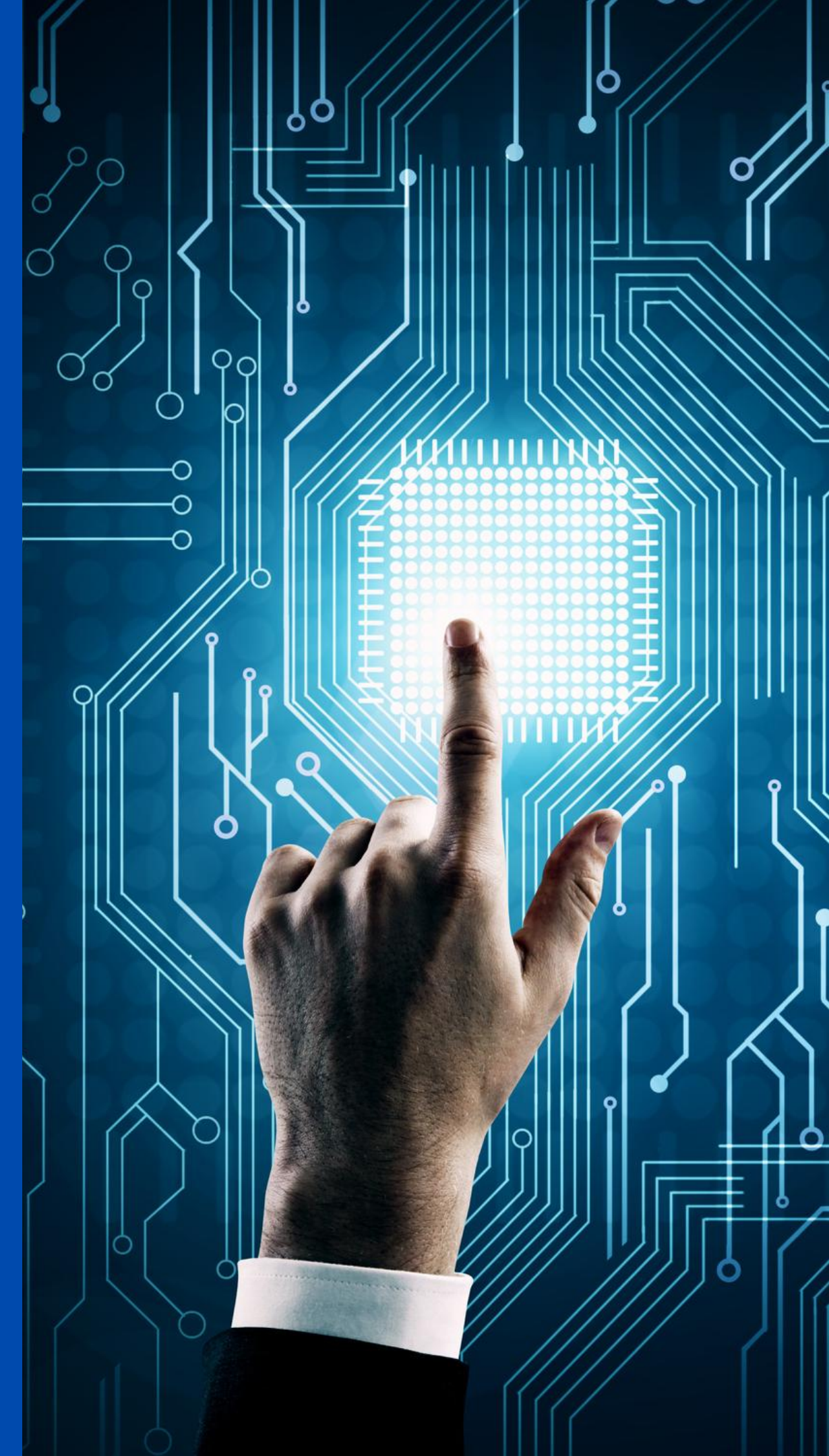
OPERATING SYSTEM

- For example, Windows 10 is an operating system that allows you to run programs like Microsoft Word, browse the internet, and play games, all while managing the computer's resources efficiently so that everything runs smoothly.



GENERAL TASK OF OPERATING SYSTEM

- Control of the operation of the input, output and backing storage.
- supervising the loading, running, and storage of application programs
- dealing with errors that occur in application programs
- maintaining the security of the whole computer system (hardware and software)
- allowing communication between user and the computer system (user interfaces)



EXAMPLES OF OS

MICROSOFT WINDOWS

- Developed by Microsoft, Windows is the most widely used desktop operating system worldwide.
- Versions: Windows 11, Windows 10, Windows 8.1, Windows 7 (older versions like Windows XP and Windows Vista are still in use but less common).

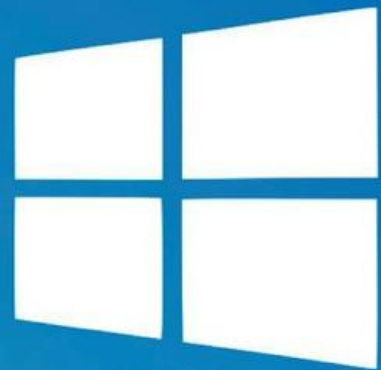
MICROSOFT WINDOWS

Evolution of Windows OS



MICROSOFT WINDOWS

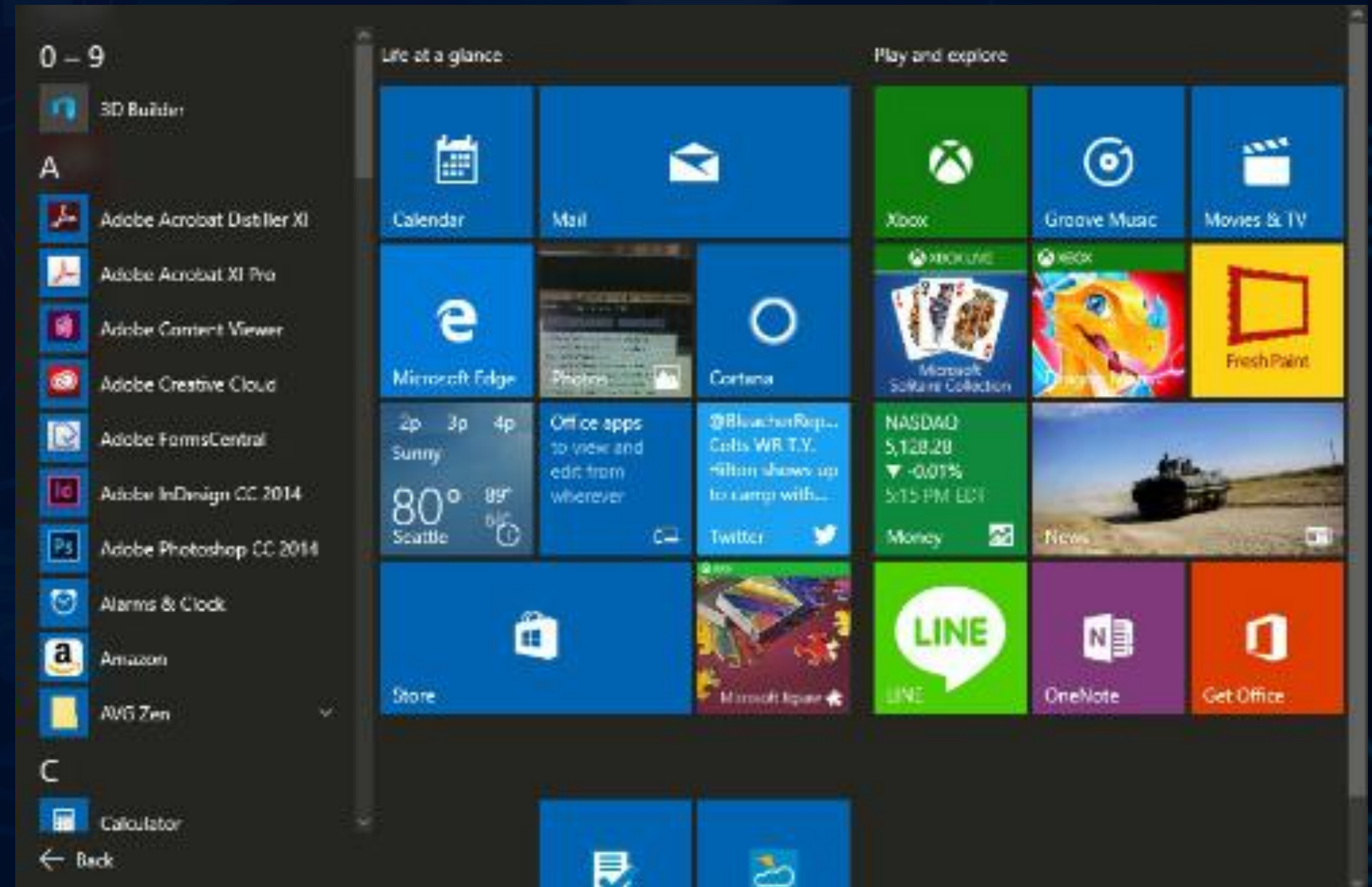
Icon Comparison



Windows 10



Windows 11



Microsoft Windows

EXAMPLES OF OS

MACOS

- Developed by Apple Inc., macOS is the operating system used exclusively on Apple Macintosh computers.
- macOS Big Sur, macOS Catalina, macOS Mojave.

MACOS



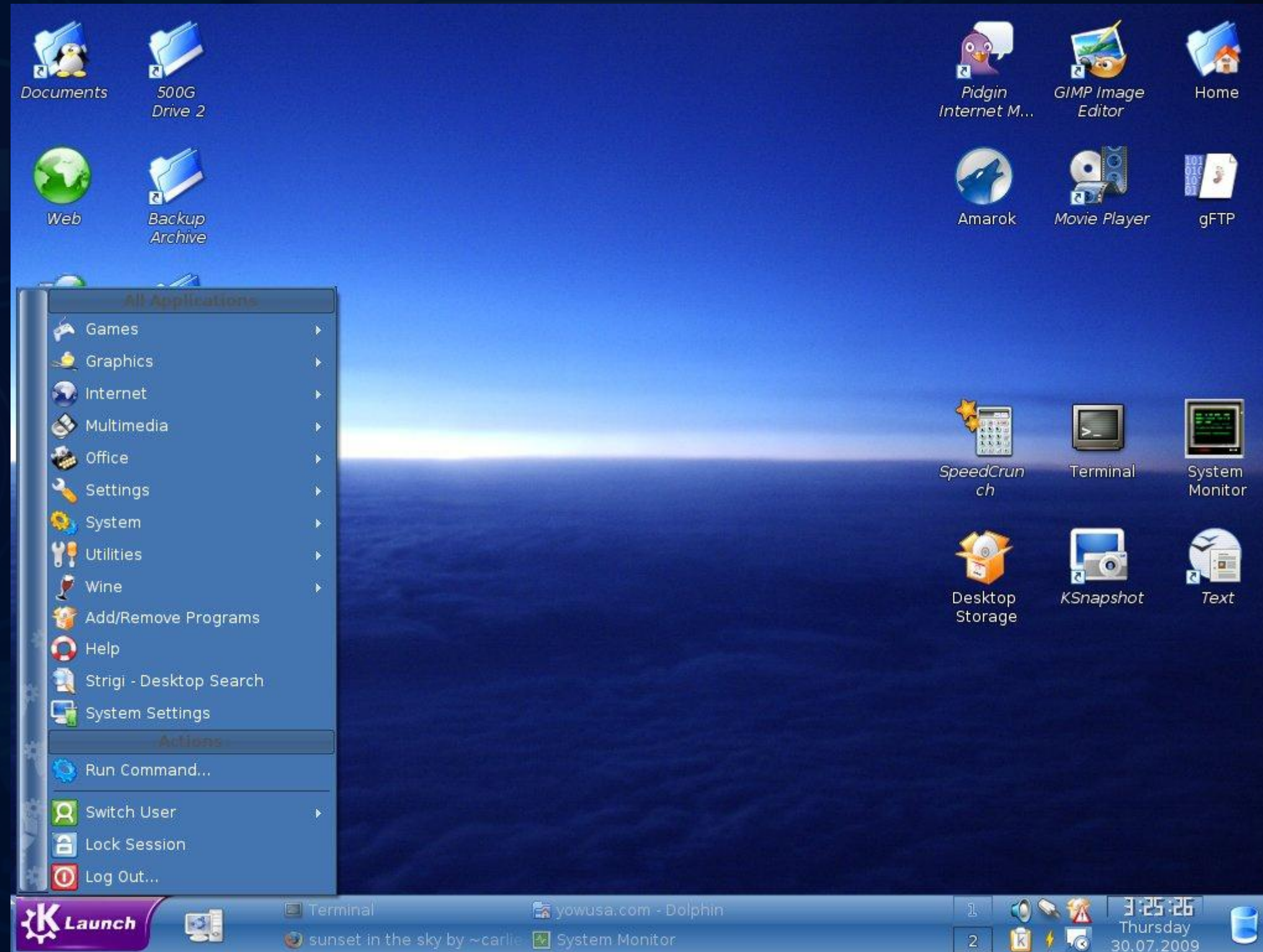
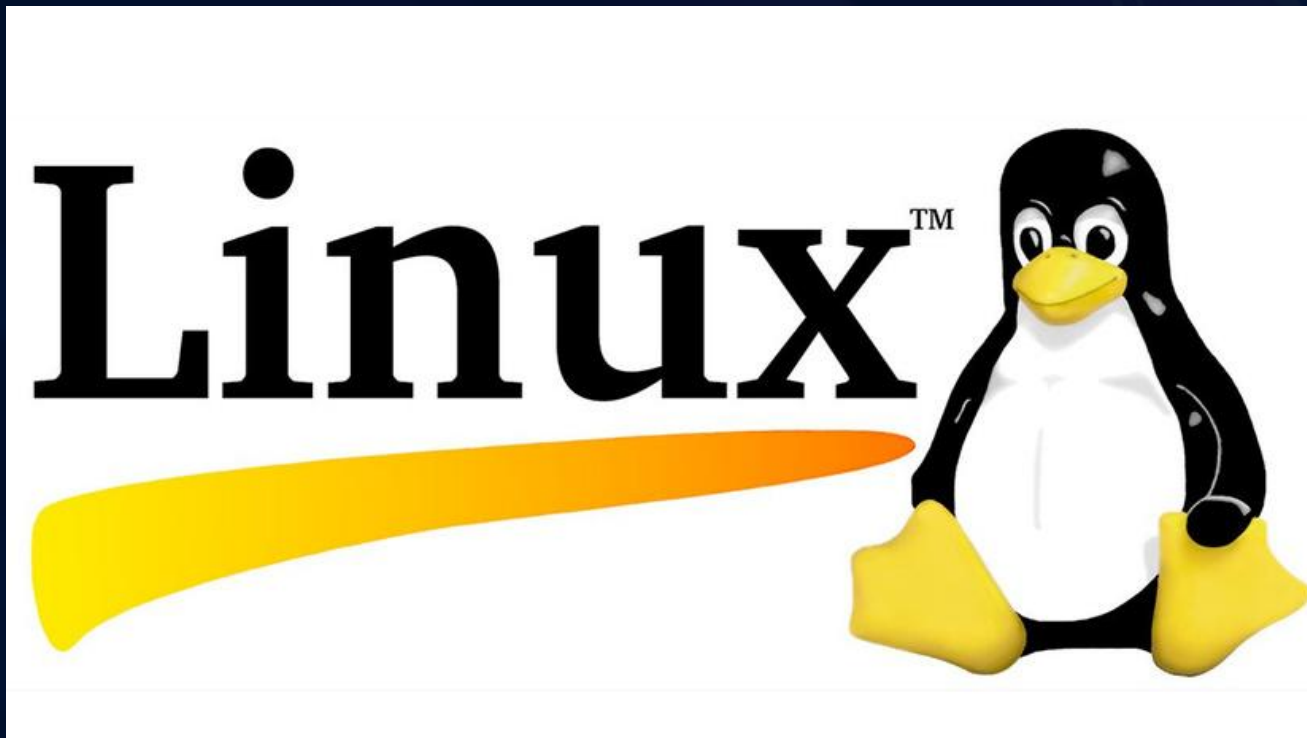
MacOS

EXAMPLES OF OS

LINUX

- Linux is an open-source operating system kernel that forms the basis of numerous Linux distributions (distros).
- Ubuntu, Fedora, Debian, CentOS, Arch Linux.

LINUX



Linux

EXAMPLES OF OS

CHROME OS

- Developed by Google, Chrome OS is a lightweight operating system designed primarily for Chromebook laptops.
- Based on the Linux kernel, focused on web applications and cloud computing, integrates with Google services, such as Gmail and Google Drive.

CHROME OS



Chrome OS

EXAMPLES OF OS

UBUNTU

- An open-source Linux distribution based on Debian, known for its user-friendly interface and extensive software repositories.
- Ubuntu Desktop (for personal computers), Ubuntu Server (for servers), Ubuntu Touch (for mobile devices).

UBUNTU



Ubuntu

UBUNTU



Ubuntu

EXAMPLES OF OS

IOS

- Developed by Apple Inc., iOS is the operating system exclusively used on iPhones, iPads, and iPod Touch devices.
- Intuitive touchscreen interface, extensive App Store with millions of applications, seamless integration with other Apple devices.

iOS



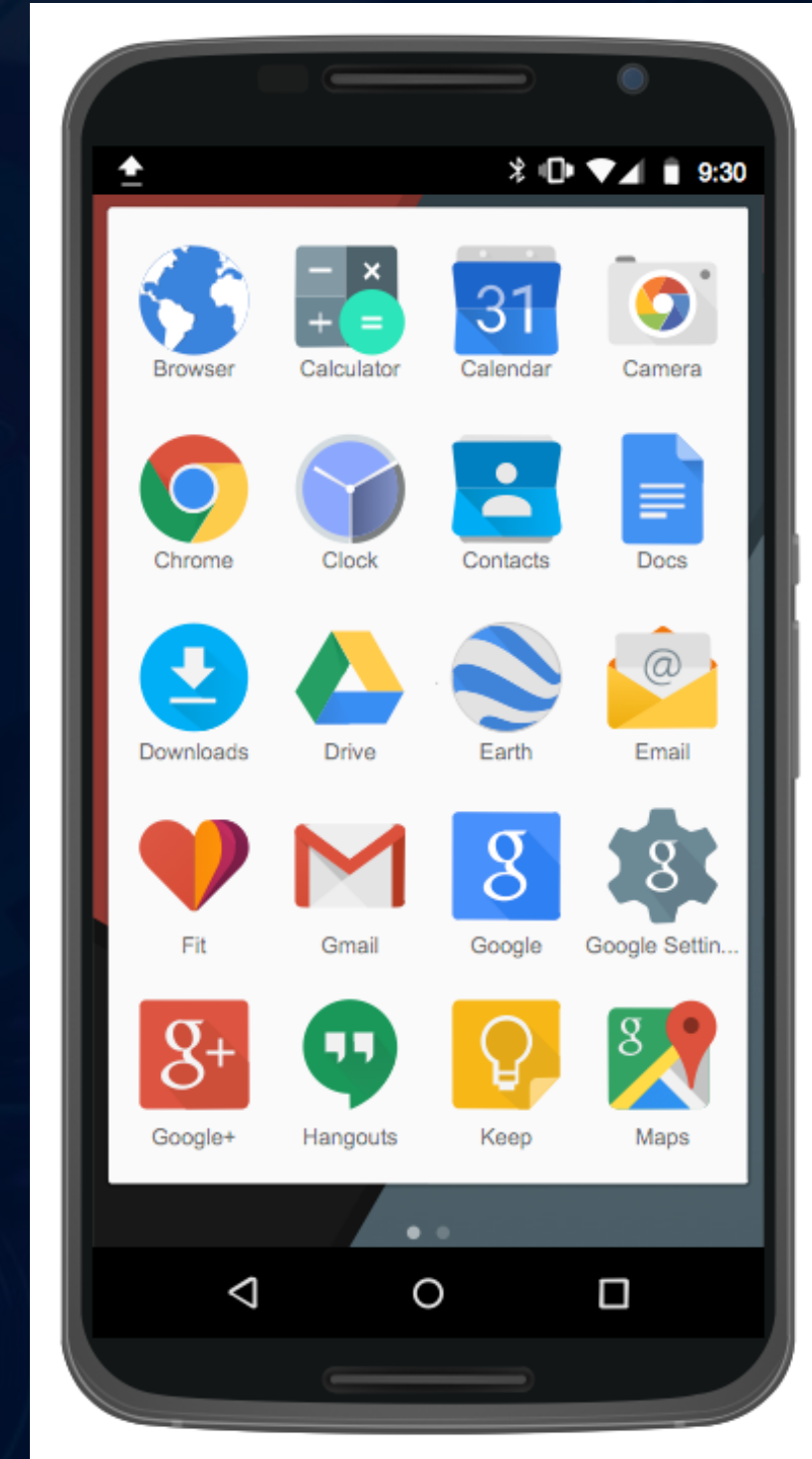
iOS

EXAMPLES OF OS

ANDROID

- Developed by Google, Android is an open-source operating system based on the Linux kernel, primarily used on smartphones, tablets, and other mobile devices.
- Customizable user interface, vast ecosystem of apps through Google Play Store, used by various manufacturers on their devices.

ANDROID



Android

IOS VS ANDROID



iOS vs. Android

IOS VS ANDROID



EXAMPLES OF OS

WINDOWS SERVER

- A version of the Windows operating system designed for server hardware, providing features tailored for networking, remote access, and server management.
- Windows Server 2019, Windows Server 2016, Windows Server 2012 R2.

WINDOWS SERVER



Windows Server

WINDOWS SERVER



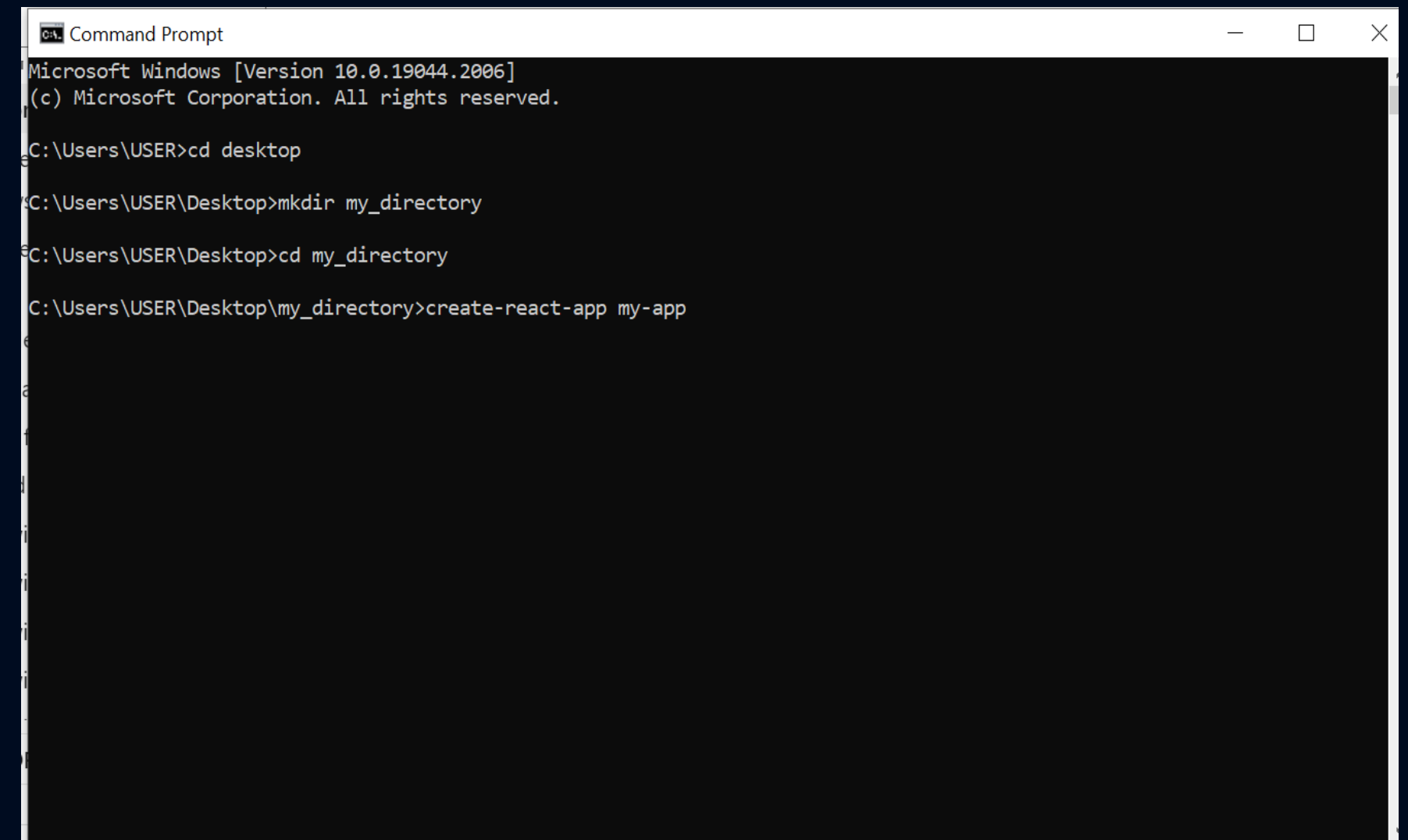
USER INTERFACE (UI)

- It is the component of the operating system or software application that allows users to interact with the computer. It provides the means through which users communicate with the system.

COMMAND LINE INTERFACE (CLI)

- relies on **text-based** commands entered through a terminal or command prompt.
- It offers more precise control over the system and is often preferred by **advanced users and system administrators** for its efficiency and flexibility in executing commands and automating tasks.

TYPES OF UI



```
C:\> Command Prompt
Microsoft Windows [Version 10.0.19044.2006]
(c) Microsoft Corporation. All rights reserved.

C:\Users\USER>cd desktop
C:\Users\USER\Desktop>mkdir my_directory
C:\Users\USER\Desktop>cd my_directory
C:\Users\USER\Desktop\my_directory>create-react-app my-app
```

GRAPHICAL USER INTERFACE (GUI)

- provides a visually intuitive environment with graphical elements like windows, icons, menus, and buttons, allowing users to interact with the system through mouse clicks and touch gestures.

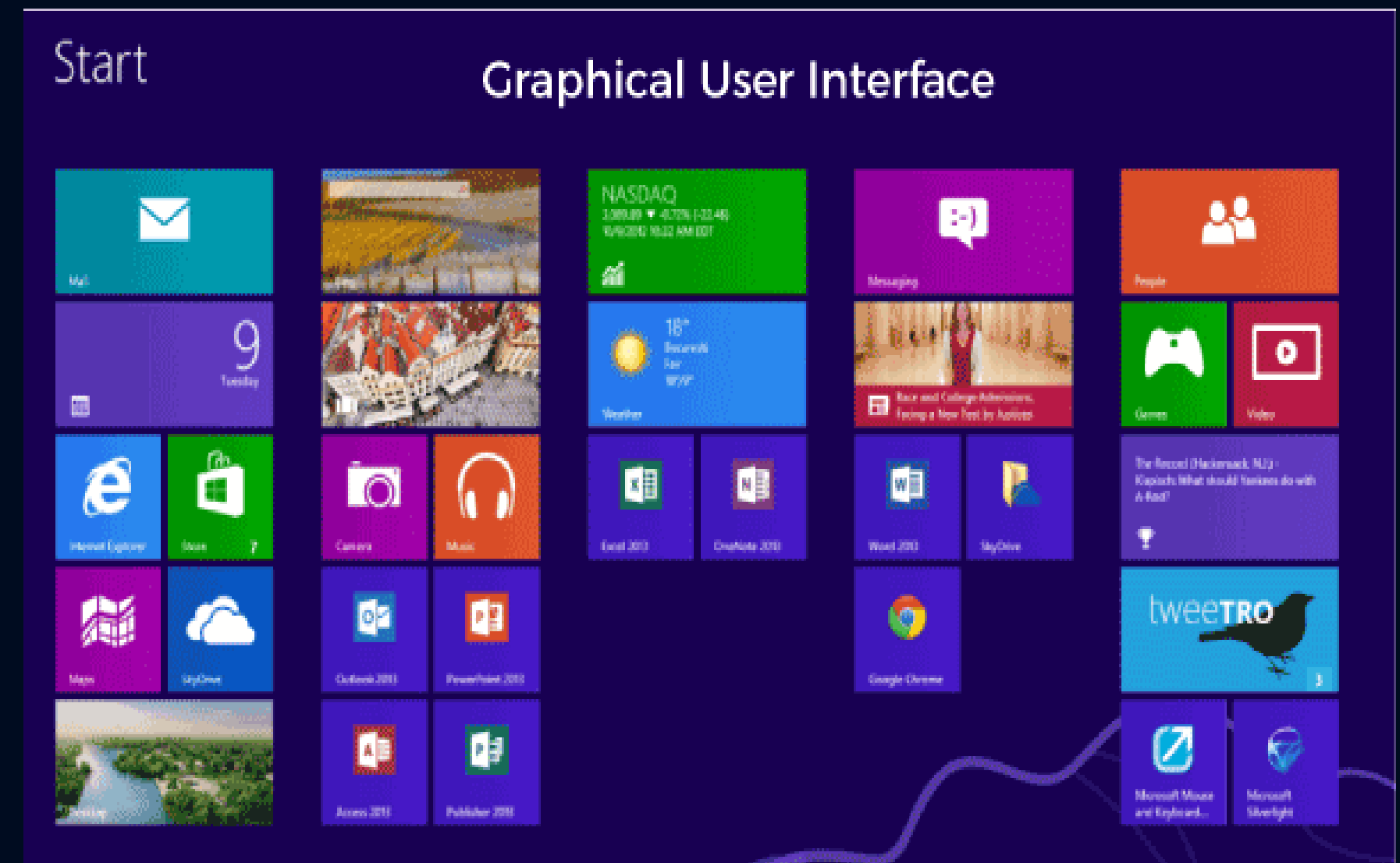
TYPES OF UI



DIALOGUE BASED INTERFACE (DBI)

- Use human voice to give command to computer.
- Instead of navigating through menus or clicking buttons, users communicate with the system using natural language, just like they would in a conversation with another person.
- **Chatbots** (websites), **VA** (Siri, Google Assistant, and Alexa)

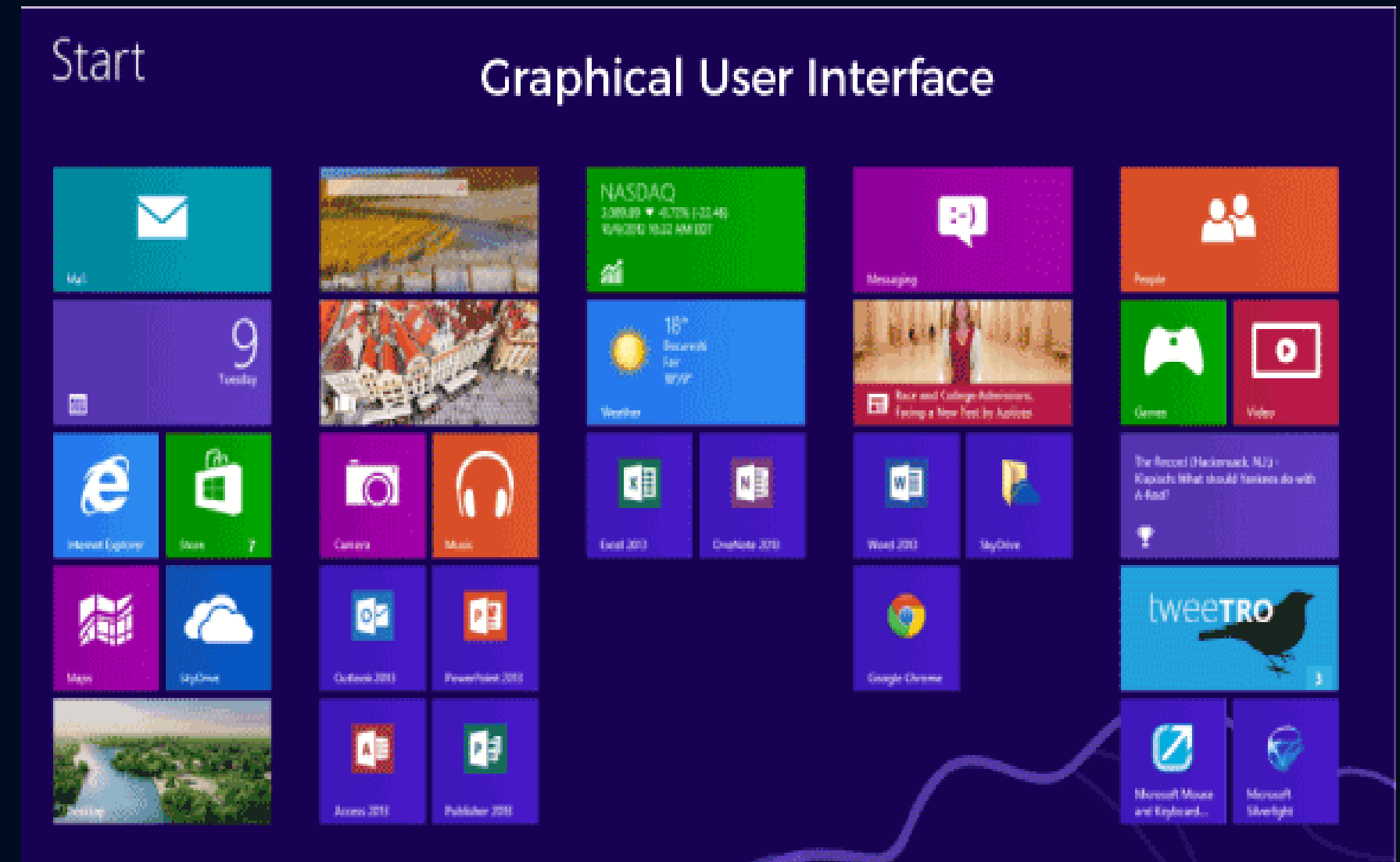
TYPES OF UI



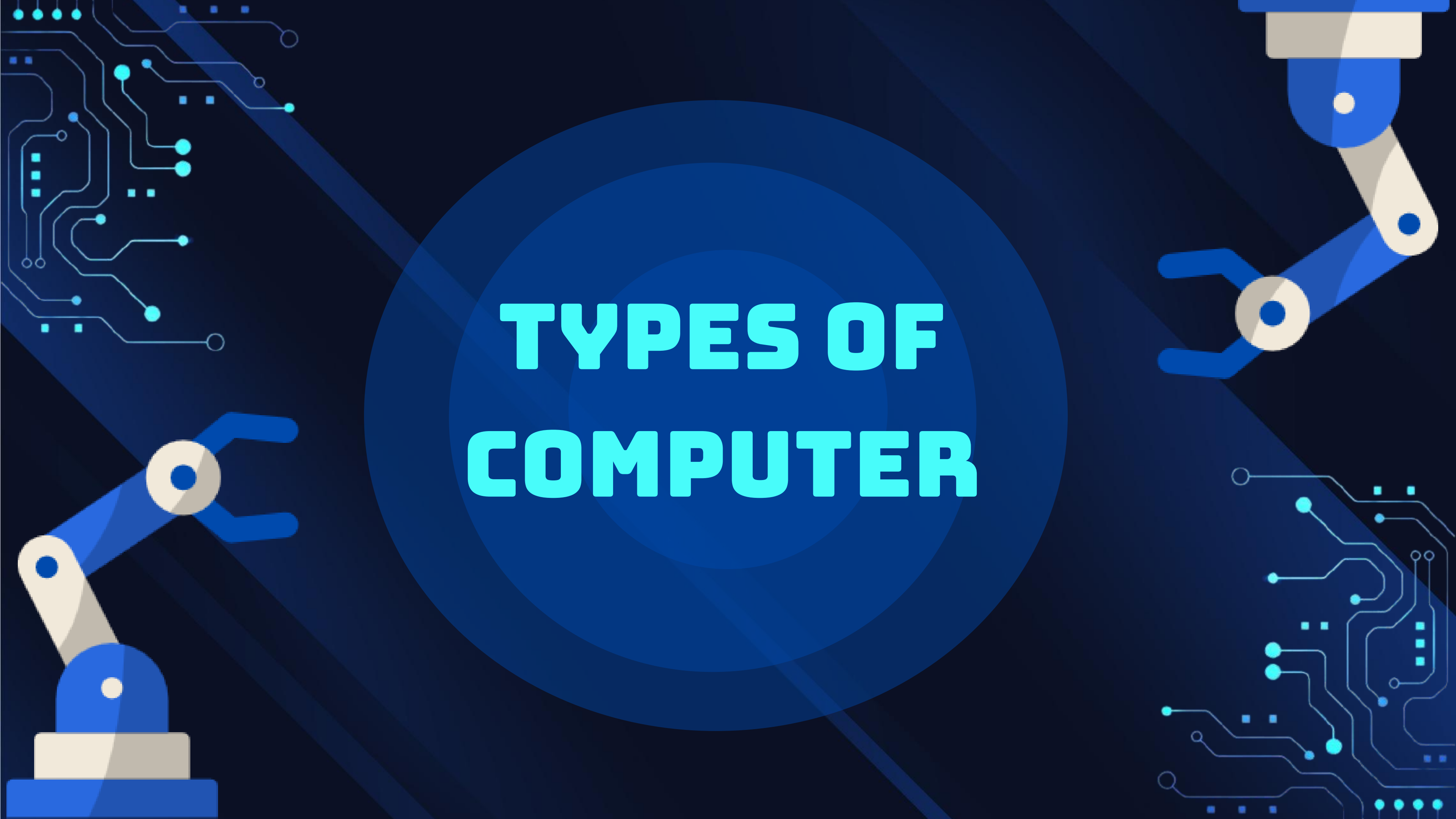
GESTURE BASED INTERFACE (GBI)

- Rely on human interaction by the moving of hands, head or even the feet.
- These gestures can be made using various parts of the body, such as hands, arms, and even facial expressions.
- Touchscreens with Multi-Touch Gestures, Virtual and Augmented Reality

TYPES OF UI



TYPES OF COMPUTER



DESKTOP COMPUTER

- It is the component of the operating system or software application that allows users to interact with the computer. It provides the means through which users communicate with the system.